

riod, and there were no significant differences in other maternal characteristics (Table 1).

The incidence of stillbirth was significantly higher during the pandemic period ($n = 16$ [9.31 per 1000 births]; none associated with COVID-19) than during the prepandemic period ($n = 4$ [2.38 per 1000 births]) (difference, 6.93 [95% CI, 1.83-12.0] per 1000 births; $P = .01$) (Table 2), and the incidence of stillbirth was significantly higher when late terminations for fetal abnormality were excluded during the pandemic period (6.98 per 1000 births vs 1.19 per 1000 births in the prepandemic period; difference, 5.79 [95% CI, 1.54-10.1]; $P = .01$). There were no significant differences over time in births before 37 weeks' gestation, births after 34 weeks' gestation, neonatal unit admission, or cesarean delivery (Table 2).

During the pandemic period, 19 patients with COVID-19 were hospitalized in the study site maternity department. None of the pregnant women who experienced stillbirth had symptoms suggestive of COVID-19, nor did the postmortem or placental examinations suggest SARS-CoV-2 infection. Universal testing for SARS-CoV-2 started on May 28, 2020, and only 1 pregnant woman, who had a live birth, had a positive test result.

Discussion | This study demonstrates an increase in the stillbirth rate during the pandemic. A direct consequence of SARS-CoV-2 infection is possible. Although none of the stillbirths in the pandemic period were among women with COVID-19, surveillance studies in pregnant women reported that as much as 90% of SARS-CoV-2-positive cases were asymptomatic.³⁻⁵ Moreover, until recently, UK national policy limited testing to symptomatic individuals requiring hospitalization. Alternatively, the increase in stillbirths may have resulted from indirect effects such as reluctance to go to the hospital when needed (eg, with reduced fetal movements), fear of contracting infection, or not wanting to add to the National Health Service burden. Changes in obstetric services may have played a role secondary to staff shortages or reduced antenatal visits, ultrasound scans, and/or screening. Although differences in the populations in the 2 periods were observed, the lower proportion of nulliparous and hypertensive women during the pandemic period would have been expected to be associated with a lower rather than higher risk of stillbirth. However, hypertension in pregnancy may have been underdiagnosed during the pandemic as women had fewer face-to-face antenatal visits. Other possible explanations include change in referral patterns with more high-risk women referred to St George's Hospital or chance due to the short time frame of the study.

Limitations of this study include its retrospective nature, single-center setting, small numbers, short time frame, and lack of information on the causes of stillbirths. Moreover, a comparable period in 2019 was not used, but this should not affect the results as there is no seasonality to stillbirths in the UK.

Asma Khalil, MD

Peter von Dadelszen, PhD

Tim Draycott, MD

Austin Ugwumadu, PhD

Pat O'Brien, MBBCh, MRCOG, MFFP

Laura Magee, PhD

Author Affiliations: Fetal Medicine Unit, St George's University of London, London, United Kingdom (Khalil, Ugwumadu); School of Life Course Sciences, King's College London, London, United Kingdom (von Dadelszen, Magee); Department of Women's Health, North Bristol NHS Trust, Westbury on Trym, United Kingdom (Draycott); Department of Women's Health, University College London Hospitals, London, United Kingdom (O'Brien).

Corresponding Author: Asma Khalil, MD, Fetal Medicine Unit, Department of Obstetrics and Gynaecology, St George's University Hospitals, NHS Foundation Trust, Blackshaw Road, London SW17 0QT, United Kingdom (akhil@sgul.ac.uk).

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Concept and design: Khalil, von Dadelszen, O'Brien, Magee.

Acquisition, analysis, or interpretation of data: Khalil, Draycott, Ugwumadu, O'Brien, Magee.

Drafting of the manuscript: Khalil, Draycott, Ugwumadu, O'Brien.

Critical revision of the manuscript for important intellectual content: All authors.

Statistical analysis: Khalil.

Administrative, technical, or material support: Khalil, von Dadelszen, Ugwumadu, Magee.

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Prevalence of Psychotropic and Opioid Prescription Fills Among Community-Dwelling Older Adults With Dementia in the US

A 2015 report highlighted the paucity of knowledge about prescribing of psychotropics to adults with dementia in community settings.¹ Prior estimates are more than 10 years old and exclude benzodiazepines and opioids.^{2,3} Given the risk of harm when prescribing such central nervous system (CNS)-active medications to older adults with dementia, more data are needed to inform research and policy.⁴ We estimated CNS-active medication prescriptions among community-dwelling older adults with dementia, identifying the most commonly prescribed medications.

Methods | We identified all fee-for-service Medicare beneficiaries aged 65 years or older with a primary or secondary diagnosis of dementia (eTable 1 in the Supplement) on a claim for a face-to-face clinical encounter between October 1, 2014, and September 30, 2015. Analysis was limited to those with



Supplemental content

when prescribing such central nervous system (CNS)-active medications to older

Table 1. Prevalence of CNS-Active Medication Prescription Fills Among Community-Dwelling Older Adults With Dementia With Fee-for-Service and Part D Medicare Coverage, Overall and by Demographic Characteristics (N = 737 839)

Characteristics	Overall, %	Any CNS-active medication ^a		Antidepressants		Opioids		Anxiolytics/sedatives-hypnotics		Antiepileptics		Antipsychotics	
		%	ARR (95% CI)	%	ARR (95% CI)	%	ARR (95% CI)	%	ARR (95% CI)	%	ARR (95% CI)	%	ARR (95% CI)
All		73.5		49.8		29.8		26.8		21.9		21.6	
Sex													
Female	66.3	75.3	1 [Reference]	52.7	1 [Reference]	30.9	1 [Reference]	28.7	1 [Reference]	21.9	1 [Reference]	22.1	1 [Reference]
Male	33.7	70.0	0.92 (0.92-0.92) ^b	44.1	0.81 (0.81-0.82) ^b	27.7	0.88 (0.87-0.89) ^b	23.3	0.80 (0.79-0.80) ^b	22.1	0.97 (0.96-0.98) ^b	20.7	0.94 (0.93-0.94) ^b
Age, y													
65-74	19.4	80.6	1.08 (1.08-1.08) ^b	57.8	1.15 (1.15-1.16) ^b	36.1	1.17 (1.16-1.18) ^b	33.6	1.26 (1.25-1.27) ^b	32.3	1.41 (1.39-1.42) ^b	26.2	1.22 (1.20-1.23) ^b
75-84	42.2	74.5	1 [Reference]	51.0	1 [Reference]	30.5	1 [Reference]	26.8	1 [Reference]	22.5	1 [Reference]	21.3	1 [Reference]
≥85	38.4	68.7	0.91 (0.91-0.92) ^b	44.5	0.85 (0.84-0.85) ^b	25.9	0.84 (0.84-0.85) ^b	23.5	0.86 (0.85-0.86) ^b	16.1	0.71 (0.71-0.72) ^b	19.7	0.92 (0.91-0.93) ^b
Race/ethnicity ^c													
Non-Hispanic White	80.5	74.6	1 [Reference]	52.4	1 [Reference]	30.3	1 [Reference]	27.7	1 [Reference]	21.7	1 [Reference]	21.7	1 [Reference]
Non-Hispanic Black	9.2	66.7	0.86 (0.85-0.86) ^b	33.6	0.61 (0.60-0.62) ^b	30.0	0.90 (0.89-0.91) ^b	18.6	0.59 (0.58-0.60) ^b	23.0	0.89 (0.88-0.91) ^b	20.7	0.83 (0.82-0.84) ^b
Hispanic	6.5	75.2	0.96 (0.95-0.96) ^b	47.7	0.86 (0.85-0.87) ^b	28.8	0.85 (0.84-0.86) ^b	31.4	0.97 (0.96-0.98) ^b	24.4	0.92 (0.91-0.94) ^b	24.1	0.93 (0.91-0.95) ^b
Other	3.8	63.1	0.82 (0.81-0.83) ^b	36.8	0.68 (0.67-0.69) ^b	22.0	0.67 (0.66-0.69) ^b	20.2	0.65 (0.64-0.67) ^b	19.8	0.79 (0.77-0.80) ^b	17.8	0.71 (0.69-0.73) ^b
Low-income subsidy eligible ^d													
No	70.3	72.2	1 [Reference]	50.0	1 [Reference]	28.2	1 [Reference]	25.2	1 [Reference]	19.8	1 [Reference]	19.9	1 [Reference]
Yes	29.7	76.5	1.08 (1.08-1.09) ^b	49.4	1.05 (1.05-1.06) ^b	33.8	1.23 (1.22-1.24) ^b	30.8	1.27 (1.26-1.28) ^b	27.0	1.36 (1.35-1.37) ^b	25.6	1.33 (1.31-1.34) ^b
Rurality ^e													
Urban	86.6	73.3	1 [Reference]	49.8	1 [Reference]	29.1	1 [Reference]	26.7	1 [Reference]	21.8	1 [Reference]	21.9	1 [Reference]
Rural	13.4	74.5	1.00 (0.99-1.00)	49.7	0.96 (0.95-0.97) ^b	34.6	1.14 (1.13-1.15) ^b	28.1	1.01 (0.99-1.02)	22.8	0.99 (0.98-1.01)	20.0	0.88 (0.87-0.90) ^b

Abbreviations: ARR, adjusted relative risk; CNS, central nervous system.

^a Medications within each class are as classified by the American Hospital Formulary Service. In a sensitivity analysis requiring 2 or more fills, the prevalences were: any CNS-active medication, 64.4%; antidepressants, 45.0%; opioids, 17.1%; anxiolytics, 19.8%; antiepileptics, 19.0%; and antipsychotics, 18.1%.

^b $P < .001$. P value for test examining if use differs from the reference group, adjusting for the other characteristics presented in the table.

^c Race/ethnicity was identified using the Research Triangle Institute race code variable from the Medicare Master Beneficiary Summary File and was included in the analysis given known racial disparities in CNS-active prescribing.⁵

^d The low-income subsidy is a Part D premium assistance program determined by the Social Security Administration based on beneficiary income and assets. For this analysis, the variable was set as "yes" if a beneficiary was eligible for or enrolled in the Part D low-income subsidy for at least 1 month during the observation period.

^e Rurality was derived using beneficiary zip code and Rural-Urban Commuting Area Codes. Urban areas comprise urban core and suburban areas; rural areas comprise large-town, small-town, and isolated rural areas.

Table 2. Top 20 CNS-Active Medications Among Community-Dwelling Older Adults With Dementia With Fee-for-Service and Part D Medicare Coverage

Rank	Generic name	Medication class	Beneficiaries prescribed medication, % (N = 737 839)	Median (IQR)	
				Fills per person, No.	Supply per person, d
1	Hydrocodone ^a	Opioid	13.5	2 (1-5)	26 (5-120)
2	Tramadol ^a	Opioid	12.1	2 (1-4)	30 (12-110)
3	Quetiapine	Antipsychotic	12.0	6 (2-11)	240 (90-360)
4	Sertraline	Antidepressant	11.3	5 (3-11)	290 (150-360)
5	Gabapentin	Antiepileptic	11.2	5 (2-9)	240 (90-359)
6	Lorazepam	Anxiolytic/sedative-hypnotic	9.6	3 (1-7)	60 (27-210)
7	Trazodone	Antidepressant	9.4	4 (2-9)	180 (61-333)
8	Mirtazapine	Antidepressant	9.4	5 (2-11)	240 (90-360)
9	Citalopram	Antidepressant	9.1	5 (3-11)	300 (150-360)
10	Escitalopram	Antidepressant	8.5	5 (3-11)	283 (143-360)
11	Alprazolam	Anxiolytic/sedative-hypnotic	7.7	4 (1-8)	110 (30-270)
12	Oxycodone ^a	Opioid	5.7	2 (1-6)	30 (6-150)
13	Risperidone	Antipsychotic	5.5	5 (2-11)	183 (61-350)
14	Valproate/divalproex	Antiepileptic	4.3	6 (3-11)	210 (90-360)
15	Duloxetine	Antidepressant	4.0	6 (3-11)	300 (140-360)
16	Clonazepam	Anxiolytic/sedative-hypnotic	3.9	5 (2-10)	187 (60-330)
17	Codeine ^a	Opioid	3.7	1 (1-2)	10 (4-30)
18	Levetiracetam	Antiepileptic	3.3	5 (3-11)	270 (120-360)
19	Zolpidem	Anxiolytic/sedative-hypnotic	3.0	3 (1-6)	90 (30-210)
20	Paroxetine	Antidepressant	2.9	5 (3-11)	300 (150-360)

Abbreviations: CNS, central nervous system; IQR, interquartile range.

^a Includes combination products containing ranked opioids (eg, hydrocodone row includes fills for hydrocodone-acetaminophen).

Medicare Part D prescription drug coverage and 100 days or fewer in a nursing home during the observation period.

We estimated overall and class-specific CNS-active prescription fills for 5 categories: antidepressants, anxiolytics/sedatives-hypnotics (“anxiolytics”), antipsychotics, opioids, and antiepileptics (eTable 2 in the [Supplement](#)).⁴ Within each class we computed prevalence of prescription fills overall and by demographic strata (Table 1). Log-binomial regression was used to estimate relative prescribing risk, adjusting for demographic characteristics. We identified the top medications prescribed and person-level volume of fills. One fill defined use; a sensitivity analysis was also performed requiring 2 or more fills during the observation period.

Analyses were conducted using SAS version 9.4 (SAS Institute Inc); tests were 2-sided with $\alpha = .05$. This study was approved by the Michigan Medicine institutional review board; informed consent was waived.

Results | We identified 737 839 community-dwelling beneficiaries with dementia; the mean age was 82.2 years and 66.3% were female. A total of 73.5% were prescribed any CNS-active medication (Table 1). Prevalence of any prescription fill was highest among beneficiaries who were female (75.3%), aged 65 to 74 years (80.6%), non-Hispanic White (74.6%), and low-income (76.5%). These same groups had the highest adjusted risks. When 2 or more fills were required, the prevalence of any CNS-active medication use was 64.4%.

A total of 49.8% filled an antidepressant prescription and 29.8% an opioid, followed by anxiolytics (26.8%), antiepileptics (21.9%), and antipsychotics (21.6%). Associations between demographics and prescription fills (both prevalent and

adjusted risk) were generally consistent in direction for individual drug classes: higher among female, younger (ie, 65-74 years), non-Hispanic White, and low-income older adults. Rurality had mixed associations with prescription fills, with higher prevalence of opioids (34.6% vs 29.1%; difference, 5.5%; 95% CI, 5.2%-5.9%) but slightly lower of antipsychotics (20.0% vs 21.9%; difference, -1.8%; 95% CI, -2.1% to -1.5%).

The opioids hydrocodone (13.5%) and tramadol (12.1%) were the top medications filled (Table 2). Per-person volumes for both were relatively low (medians of 2 prescriptions and 26 days for hydrocodone and 2 prescriptions and 30 days for tramadol). They were followed by a medication from each of the remaining classes: quetiapine (12.0% [antipsychotic]), sertraline (11.3% [antidepressant]), gabapentin (11.2% [antiepileptic]), and lorazepam (9.6% [anxiolytic]).

Discussion | Prescribing of CNS-active medications to community-dwelling Medicare beneficiaries with dementia was frequent, raising concerns about the risk-benefit balance. Opioid prescribing to these patients was common but at relatively low volumes. In contrast, gabapentin and antidepressants were prescribed in larger quantities. There was significant variation in prescription fills by patient demographics.

The analysis reflects prescribing to fee-for-service beneficiaries from 2015 to 2016, which may not generalize to all older adults in 2020. Other limitations include no information available on prescriptions’ appropriateness, and that the data reflect prescription fills as a proxy for use. However, class-specific fills among this cohort with dementia far exceeded that by older adults in the general population in which, for example, antidepressant and antipsychotic use was found among

17.3% and 1.4% of older adults, respectively, in 2013.⁵ Compared with prior studies of community-dwelling persons with dementia,^{2,3} the higher overall prevalence of prescription fills is potentially related to the broader scope of CNS-active prescribing captured. Rates found in this study are similar to long-term care settings, in which such prescribing has received significant regulatory attention.⁶ None of these medications are approved in the US to treat dementia or its behavioral symptoms; all are associated with adverse events among older adults, including falls, sedation, and increased risk of death.⁴ Work is needed to understand factors driving prescribing given scant evidence of benefit for adults with dementia.

Donovan T. Maust, MD, MS
Julie Strominger, MS
Julie P. W. Bynum, MD
Kenneth M. Langa, MD, PhD
Lauren B. Gerlach, DO, MS
Kara Zivin, PhD
Steven C. Marcus, PhD

Author Affiliations: Department of Psychiatry, University of Michigan, Ann Arbor (Maust, Gerlach, Zivin); Center for Clinical Management Research, VA Ann Arbor Healthcare System, Ann Arbor, Michigan (Strominger); Department of Medicine, University of Michigan, Ann Arbor (Bynum, Langa); University of Pennsylvania School for Social Policy and Practice, Philadelphia (Marcus).

Corresponding Author: Donovan T. Maust, MD, MS, Department of Psychiatry, University of Michigan, 2800 Plymouth Rd, NCRC 016-226W, Ann Arbor, MI 48109 (maustd@umich.edu).

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Author Contributions: Dr Maust had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Maust, Strominger, Bynum, Gerlach, Marcus.

Acquisition, analysis, or interpretation of data: Maust, Strominger, Langa, Gerlach, Zivin, Marcus.

Drafting of the manuscript: Maust, Strominger, Gerlach, Marcus.

Critical revision of the manuscript for important intellectual content: Maust, Bynum, Langa, Gerlach, Zivin, Marcus.

Statistical analysis: Strominger, Marcus.

Obtained funding: Maust.

Administrative, technical, or material support: Maust, Strominger, Zivin.

Supervision: Maust, Bynum.

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COMMENT & RESPONSE

Uncemented vs Cemented Hemiarthroplasty for Hip Fracture

To the Editor Dr Okike and colleagues found that uncemented hemiarthroplasty, compared with cemented fixation, was associated with a higher risk of aseptic revision.¹ We have several concerns.

First, the possibility of confounding by other factors should be considered when assessing the association between an exposure and a patient outcome in a retrospective cohort study. Nevertheless, several confounders were not adjusted in this study. The surgical approach of hemiarthroplasty can influence outcomes in hip surgery. The anterior approach, compared with the lateral or posterior approach, has been associated with a significantly higher risk of revision in patients undergoing total hip arthroplasty.² Additionally, other factors, such as implant type and postoperative functional exercise, could also influence the outcomes.

Second, considering that infection has been recognized as one of the main causes of prosthetic revision, it seems inappropriate to regard aseptic revision as the primary outcome measure. As to the secondary outcomes, several studies have found that cemented hemiarthroplasty was associated with significantly increased mortality within the first postoperative day compared with uncemented hemiarthroplasty.^{3,4} Early postoperative mortality was not assessed in this study.

Third, the median length of follow-up was 3.8 years after surgery. Given that the implant survivorship in hip arthroplasty is generally more than 10 years, such a short-term follow-up could not reveal the ultimate outcomes of the procedures.

Fourth, the authors suggested that US surgeons should choose cemented over uncemented hemiarthroplasty in hip fracture. Nevertheless, aside from the revision rates, many other factors, such as postsurgery function recovery, postsurgery pain perception, and cost-effectiveness, might also influence clinical decisions. Furthermore, hemiarthroplasty, as a complex procedure, requires frequent repetition for high-quality performance. Adverse outcomes are more likely to happen during a learning curve. The instrumental variable analysis in this study showed a 0.3% to 0.4% decrease in revision rate compared with the main analysis, indicating that experienced surgeons could mitigate some of the complications compared with the overall surgeon group. Conceivably, the reduction could be even greater if compared with inexperienced surgeons. Thus, surgeons should choose the fixation technique with which they are most familiar to avoid errors caused by a learning curve.

Yang Yu, MD, PhD
Shitao Lu, MD, PhD
Jianzhong Xu, MD